

5G RAN

Modulation Schemes Feature Parameter Description

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1 Change History

The following describes the specific changes in different issues.

5G RAN10.1 Draft A (2025-12-31)

This issue introduces the following changes to 5G RAN9.1 03 (2025-08-30).

- Technical changes

Change Description	Parameter Change	RAT	Base Station Model
Added an impact relationship between DL 256QAM and L/NR Inter-band Deep Coordination. For details, see 5.3.2.3 Function Impacts .	None	NR TDD	3900 and 5900 series base stations (macro base stations)
Added an impact relationship between DL 1024QAM and L/NR Inter-band Deep Coordination. For details, see 6.3.2.3 Function Impacts .	None	NR TDD	3900 and 5900 series base stations (macro base stations)

- Editorial changes

Added the statement that LampSite base stations do not support DL 256QAM in high-frequency TDD. For details, see [5.3.3 Hardware](#).

Added descriptions of the mutually exclusive relationship between cell deployment in IAB mode and UL 256QAM. For details, see [8.3.2.2 Mutually Exclusive Functions](#).

Revised descriptions in this document.

2 About This Document

2.1 General Statements

Purpose

Feature Parameter Description documents are intended to acquaint readers with:

- The technical principles of features and their related parameters
- The scenarios where these features are used, the benefits they provide, and the impact they have on networks and functions
- Requirements that must be met before feature activation
- Parameter configuration required for feature activation, verification of feature activation, and monitoring of feature performance

NOTE

This document only provides guidance for feature activation. Feature deployment and feature gains depend on the specifics of the network scenario where the feature is deployed. To achieve optimal gains, contact Huawei professional service engineers.

Functions mentioned in this document work properly only when enabled in the specified applicable scenarios (such as RAT and networking). If a function not mentioned in this document is enabled or a function is enabled in a scenario not specified as applicable, exceptions or other impacts may occur.

Software Interfaces

Any parameters, alarms, counters, or managed objects (MOs) described in Feature Parameter Description documents apply only to the corresponding software release. For future software releases, refer to the corresponding updated product documentation.

2.2 Features in This Document

This document describes the following features.

Feature ID	Feature Name	Chapter/Section
FBFD-010008	Basic Modulation Schemes	4 Basic Modulation Schemes
FOFD-010050	DL 256QAM	5 DL 256QAM 7 Downlink MCS Index Table Adaptation
FOFD-081237	DL 1024QAM	6 DL 1024QAM 7 Downlink MCS Index Table Adaptation
FOFD-030215	UL 256QAM	8 UL 256QAM

2.3 Differences

Differences Between NSA and SA

Function Name	Difference	Chapter/Section
Basic modulation schemes	None	4 Basic Modulation Schemes
DL 256QAM	None	5 DL 256QAM
DL 1024QAM	None	6 DL 1024QAM
Downlink MCS index table adaptation	None	7 Downlink MCS Index Table Adaptation
UL 256QAM	None	8 UL 256QAM

Differences Between High Frequency Bands and Low Frequency Bands

This document refers to frequency bands belonging to FR1 (410–7125 MHz) as low frequency bands, and those belonging to FR2 (24250–71000 MHz) as high frequency bands. For details about FR1 and FR2, see section 5.1 "General" in 3GPP TS 38.104 V17.8.0.

Function Name	Difference	Chapter/Section
Basic modulation schemes	Supported in both high and low frequency bands with no differences	4 Basic Modulation Schemes
DL 256QAM	Supported in both high and low frequency bands but with different supported hardware	5 DL 256QAM

Function Name	Difference	Chapter/Section
DL 1024QAM	Supported only in low frequency bands	6 DL 1024QAM
Downlink MCS index table adaptation	Supported in both low and high frequency bands, with the following differences: • "Adaptation between DL 256QAM MCS index table and DL 64QAM MCS index table" is supported in both high and low frequency bands, but the supported hardware differs. • "Adaptation between DL 1024QAM MCS index table and DL 256QAM MCS index table and between DL 256QAM MCS index table and DL 64QAM MCS index table" is supported only in low frequency bands.	7 Downlink MCS Index Table Adaptation
UL 256QAM	Supported only in low frequency bands	8 UL 256QAM

3 Overview

3.1 Modulation Schemes

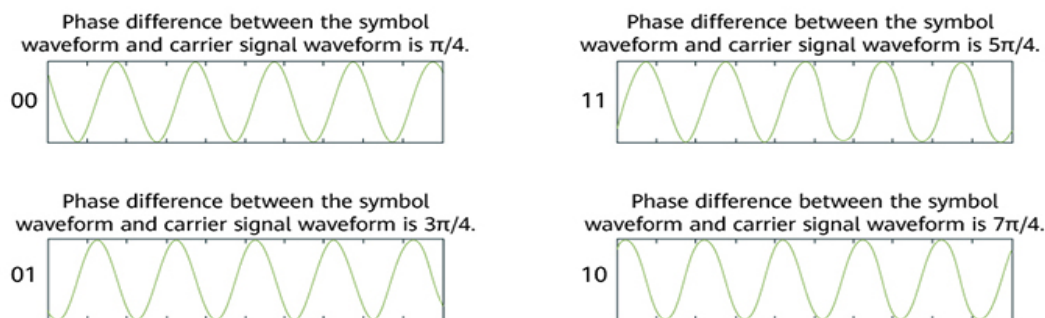
In digital communications, modulation is the process of adding information into carriers for transmission. It involves varying the amplitude, phase, or frequency of carriers in accordance with digital baseband signals. Common modulation schemes include the following:

- Phase shift keying (PSK): PSK varies the phase of a carrier in accordance with a digital signal, while keeping the amplitude and frequency unchanged. For details, see [PSK](#).
- Quadrature amplitude modulation (QAM): QAM combines amplitude and phase modulation. It varies the amplitudes of two carriers of the same frequency and in quadrature (90° out of phase) with each other. For details, see [QAM](#).

Modulation schemes may vary in modulation orders. For details about modulation orders, see [Modulation Order](#).

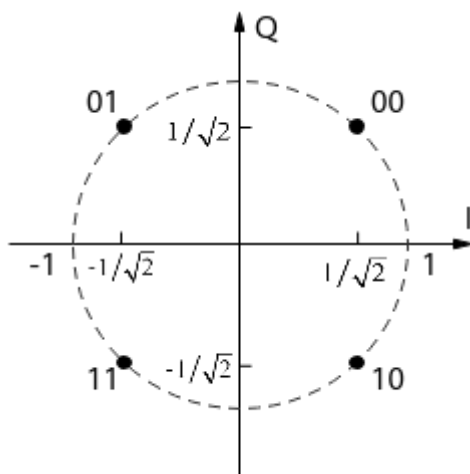
PSK

Quadrature phase shift keying (QPSK) is a common form of PSK and is used as an example to illustrate PSK in this document. QPSK conveys a digital signal by varying the phase of a carrier using four distinct phase states. That is, four symbol waveforms with different initial phases are used to represent different data, as shown in [Figure 3-1](#), and each symbol waveform represents a unique two-bit code (00, 01, 11, or 10).

Figure 3-1 Four symbol waveforms in QPSK

A constellation diagram visualizes how input data is mapped onto in-phase/quadrature (I/Q) data and carrier phases for a given modulation scheme. I/Q data is the coordinates of a constellation point in an I/Q coordinate system (which is a complex coordinate system). Each constellation point represents a modulation symbol. The amplitude and phase of the vector from the origin to a constellation point represents the amplitude shift and phase shift applied to the carrier for that modulation symbol.

Figure 3-2 shows the constellation diagram for QPSK.

Figure 3-2 Constellation diagram for QPSK

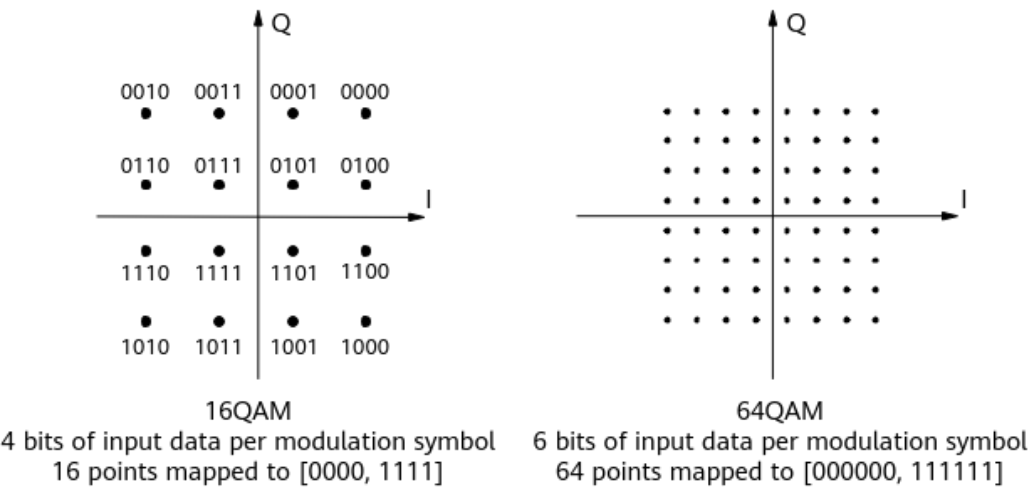
QAM

QAM combines amplitude and phase modulation. It varies the amplitudes of two carriers of the same frequency and in quadrature with each other, and generates transmitted signals by adding the two modulated carriers together. QAM forms include 16QAM, 64QAM, 256QAM, and 1024QAM, which differ in terms of their modulation order.

In a QAM constellation diagram, the points do not lie on the unit circle because QAM varies both the amplitude and phase of carriers, while PSK only varies the phase. Instead, the points for QAM are scattered in certain patterns, with each form of QAM having a unique pattern. In the constellation diagram for a form of QAM, each point corresponds to a unique combination of amplitude and phase.

Figure 3-3 shows the constellation diagrams for 16QAM and 64QAM.

Figure 3-3 Constellation diagrams for 16QAM and 64QAM



Modulation Order

The modulation order is represented by the number of bits per modulation symbol, which equals $\log_2 N$, with N being the number of modulation symbols involved in a modulation scheme. A larger value of $\log_2 N$ indicates a higher modulation order. Table 3-1 lists the modulation orders of different modulation schemes.

Table 3-1 Modulation orders of different modulation schemes

Modulation Scheme	Number of Modulation Symbols (N)	Modulation Order ($\log_2 N$)
QPSK	4	2
16QAM	16	4
64QAM	64	6
256QAM	256	8
1024QAM	1024	10

3.2 MCS

In a New Radio (NR) communications system, a modulation and coding scheme (MCS) is a combination of a modulation scheme and a coding rate. The gNodeB selects an MCS for a UE that both the gNodeB and UE support. To balance transmission efficiency and quality, an MCS is selected based on the radio channel quality:

- When a UE is in a radio environment with favorable channel quality, the gNodeB selects an MCS with a higher-order modulation scheme. Such an MCS results in a higher transmission efficiency but also a higher block error rate (BLER) given the same radio environment.

- When a UE is in a radio environment with poor channel quality, the gNodeB selects an MCS with a lower-order modulation scheme. Such an MCS results in a lower transmission efficiency but also a lower BLER given the same radio environment.

Each combination of a modulation scheme and a coding rate is identified by an MCS index in a 3GPP-defined MCS index table. An MCS index table defines the modulation order Q_m , target coding rate, and spectral efficiency corresponding to each MCS index. The following MCS index tables are defined, which are named according to the highest-order modulation scheme in a table:

- DL 64QAM MCS index table: The highest-order downlink modulation scheme in the table is 64QAM.
- DL 256QAM MCS index table: The highest-order downlink modulation scheme in the table is 256QAM.
- DL 1024QAM MCS index table: The highest-order downlink modulation scheme in the table is 1024QAM.
- UL 64QAM MCS index table: The highest-order uplink modulation scheme in the table is 64QAM.
- UL 256QAM MCS index table: The highest-order uplink modulation scheme in the table is 256QAM.

 NOTE

For details about MCS selection by the gNodeB, see *Downlink Scheduling* and *Uplink Scheduling*.

For details about MCS index tables, see sections 5.1.3.1 "Modulation order and target code rate determination" and 6.1.4.1 "Modulation order and target code rate determination" in 3GPP TS 38.214.

3.3 Functions Related to Modulation Schemes

The modulation schemes described in this document comply with section 5.1 "Modulation mapper" in 3GPP TS 38.211 V17.4.0. Huawei supports the following modulation schemes:

- Uplink modulation schemes
 - UL QPSK
 - UL 16QAM
 - UL 64QAM
 - UL 256QAM
- Downlink modulation schemes
 - DL QPSK
 - DL 16QAM
 - DL 64QAM
 - DL 256QAM
 - DL 1024QAM

This document describes the following functions:

- Basic modulation schemes, which refer to 64QAM and lower-order modulation schemes. For details, see [4 Basic Modulation Schemes](#).
- High-order modulation schemes, which refer to 256QAM and higher-order modulation schemes and include the following:
 - DL 256QAM. For details, see [5 DL 256QAM](#).
 - DL 1024QAM. For details, see [6 DL 1024QAM](#).
 - UL 256QAM. For details, see [8.1.1 UL 256QAM](#).
- MCS index table adaptation:

MCS index table adaptation is supported for high-order modulation schemes. This function enables the optimal MCS index table to be selected adaptively based on channel quality changes to improve user experience. This function includes the following:

 - Downlink MCS index table adaptation. For details, see [7 Downlink MCS Index Table Adaptation](#).
 - Uplink MCS index table adaptation. For details, see [8.1.2 Uplink MCS Index Table Adaptation](#).

4 Basic Modulation Schemes

4.1 Principles

UL/DL QPSK, UL/DL 16QAM, and UL/DL 64QAM are all basic modulation schemes. The gNodeB selects an appropriate uplink or downlink modulation scheme for a UE based on the uplink or downlink channel quality of the UE during scheduling.

UL/DL QPSK, UL/DL 16QAM, and UL/DL 64QAM are not under switch control and are enabled by default.

4.2 Network Analysis

4.2.1 Benefits

UL/DL QPSK, UL/DL 16QAM, and UL/DL 64QAM are all basic functions and are used as the baseline for evaluating the performance of other modulation schemes.

4.2.2 Impacts

The impacts between this function and other functions are described in [4.3.2.3 Function Impacts](#).

Higher-order modulation schemes deliver higher spectral efficiency and boost system capacity. However, they are more complicated to implement, provide narrower coverage, and are less resistant to interference.

4.3 Requirements

4.3.1 Licenses

There are no license requirements for basic functions.

4.3.2 Functions

Before activating this function, ensure that its prerequisite functions have been activated and mutually exclusive functions have been deactivated, if there are any. For detailed operations, see the relevant feature documents.

Additionally, understand any potential function impacts before activating this function. To deactivate any impacted functions, follow the operations described in the relevant feature documents.

4.3.2.1 Prerequisite Functions

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	None	None	None	None
High-frequency TDD	None	None	None	None

4.3.2.2 Mutually Exclusive Functions

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	None	None	None	None
High-frequency TDD	None	None	None	None

4.3.2.3 Function Impacts

None

4.3.3 Hardware

For details about the compatibility between the hardware described in this section, see the technical specifications and hardware description of the corresponding hardware in *3900 & 5900 Series Base Station Product Documentation*.

Base Station Models

3900 and 5900 series base stations (macro base stations). 3900 series base stations must be configured with the BBU3910.

DBS3900 LampSite and DBS5900 LampSite. DBS3900 LampSite must be configured with the BBU3910.

Boards

All NR-capable main control boards and baseband processing units support this function. For details, see the BBU technical specifications in *3900 & 5900 Series Base Station Product Documentation*.

RF Modules

This function does not depend on RF modules.

4.3.4 Others

There are no other requirements for UL/DL QPSK, UL/DL 16QAM, and UL/DL 64QAM.

4.4 Operation and Maintenance

4.4.1 Data Configuration

This feature is a basic feature. It is enabled by default and does not require data configuration.

4.4.2 Activation Verification

Observe the values of the counters related to basic modulation schemes. If any of the counters related to a modulation scheme returns a non-zero value, the modulation scheme has taken effect.

- Counters related to UL QPSK
 - **N.UL.SCH.QPSK.TB**
 - **N.UL.SCH.QPSK.TB.Retrans**
 - **N.UL.SCH.QPSK.ErrTB.Rbler**
- Counters related to DL QPSK
 - **N.DL.SCH.QPSK.TB**
 - **N.DL.SCH.QPSK.TB.Retrans**
 - **N.DL.SCH.QPSK.ErrTB.Rbler**
- Counters related to UL 16QAM
 - **N.UL.SCH.16QAM.TB**
 - **N.UL.SCH.16QAM.TB.Retrans**
 - **N.UL.SCH.16QAM.ErrTB.Rbler**
- Counters related to DL 16QAM
 - **N.DL.SCH.16QAM.TB**
 - **N.DL.SCH.16QAM.TB.Retrans**
 - **N.DL.SCH.16QAM.ErrTB.Rbler**

- Counters related to UL 64QAM
 - **N.UL.SCH.64QAM.TB**
 - **N.UL.SCH.64QAM.TB.Retrans**
 - **N.UL.SCH.64QAM.ErrTB.Rbler**
- Counters related to DL 64QAM
 - **N.DL.SCH.64QAM.TB**
 - **N.DL.SCH.64QAM.TB.Retrans**
 - **N.DL.SCH.64QAM.ErrTB.Rbler**

4.4.3 Network Monitoring

This function is a basic function and takes effect by default. Network monitoring is not required.

5 DL 256QAM

5.1 Principles

5.1.1 DL 256QAM

DL 256QAM results in a higher spectral efficiency than basic downlink modulation schemes, and therefore delivers a better downlink user experience.

After DL 256QAM is enabled, the gNodeB automatically uses the DL 256QAM MCS index table for UEs that support DL 256QAM. An appropriate modulation scheme can then be selected from this table according to the downlink channel quality of each UE:

- If the downlink channel quality of a UE is favorable, DL 256QAM can be used for the UE.
- If the downlink channel quality of a UE is poor, a modulation scheme with a lower order than DL 256QAM is used for the UE.

DL 256QAM is controlled by the **NRDUCellAlgoSwitch.DL256QamSwitch** parameter.

- Once the value of this parameter is changed from **OFF** to **ON**, the DL 256QAM MCS index table is immediately used for newly admitted UEs but is not immediately used for existing online UEs.
- Once the value of this parameter is changed from **ON** to **OFF**, the DL 256QAM MCS index table is forbidden for newly admitted UEs but is not immediately changed to another table for existing online UEs for which this table is being used.

NOTE

For details about modulation schemes, see [3.1 Modulation Schemes](#). For details about the DL 256QAM MCS index table, see [3.2 MCS](#).

5.1.2 Adaptation Between DL 256QAM MCS Index Table and DL 64QAM MCS Index Table

Downlink MCS index table adaptation is supported for high-order modulation schemes. This function enables the optimal MCS index table to be selected adaptively based on channel quality changes to improve user experience.

Downlink MCS index table adaptation is controlled by the **DL_MCS_TABLE_ADAPT_SW** option of the **NRDUCellPdsch.DLinkAdaptAlgoSwitch** parameter. It includes "adaptation between DL 256QAM MCS index table and DL 64QAM MCS index table", which is detailed in [7 Downlink MCS Index Table Adaptation](#).

5.2 Network Analysis

5.2.1 Benefits

Compared with downlink basic modulation schemes, DL 256QAM improves downlink spectral efficiency for cell center users (CCUs), and under the same channel conditions, provides the following benefits:

- Given more traffic volume, the average downlink throughput of CCUs and the average downlink cell throughput increase, and downlink data packets in the radio link control (RLC) buffer wait for a shorter time to be scheduled.
- Given the same or lower traffic volume, the downlink PRB usage decreases.

The degree of benefits from DL 256QAM varies according to channel conditions. When channel conditions are favorable (for example, when the proportion of occasions where index 28 in the DL 64QAM MCS index table is selected exceeds 30% within 24 hours), DL 256QAM can increase the **Cell Downlink Average Throughput (DU)** and **User Downlink Average Throughput (DU)** by up to 30%. Otherwise, the benefits are smaller.

NOTE

Proportion of occasions where index 28 in the DL 64QAM MCS index table is selected = $\frac{\text{N.ChMeas.PDSCH.MCS.28}}{\text{Sum of N.ChMeas.PDSCH.MCS.0 to N.ChMeas.PDSCH.MCS.28}}$

5.2.2 Impacts

The impacts between this function and other functions are described in [5.3.2.3 Function Impacts](#).

This function has the following network impacts:

- Higher-order modulation schemes deliver higher spectral efficiency and boost system capacity. However, they are more complicated to implement, provide narrower coverage, and are less resistant to interference while leading to slightly increased retransmission rates.
- Given the same MCS index, more data can be transmitted by the gNodeB with the DL 256QAM MCS index table than with the DL 64QAM MCS index table in cell-center areas. Therefore, given the same amount of data to be

transmitted, the average MCS index will decrease slightly after DL 256QAM is enabled.

- Depending on UE's weight types and channel conditions, the average number of layers when DL 256QAM is used is likely to be less than that when DL 64QAM is used.
- Depending on UE's service characteristics (such as the interval and size of data packets), the CCE usage (**N.CCE.Used.Avg/N.CCE.Avail.Avg**) and number of scheduling times when DL 256QAM is used will differ from those when DL 64QAM is used.
- After DL 256QAM is enabled in high-frequency TDD, the **Cell Downlink Average Throughput (DU)** and **User Downlink Average Throughput (DU)** may decrease when CCUs and cell edge users (CEUs) are scheduled simultaneously on the same beam.

5.3 Requirements

5.3.1 Licenses

Feature ID	Feature Name	Model	Sales Unit
FOFD-010050	DL 256QAM	NR0S0DUQAM00	Per cell

NOTE

The insufficiency of certain feature licenses affects cell activation. For details, see the **Is Cell Activation Affected by License Insufficiency** column in *License Control Item Lists*.

5.3.2 Functions

Before activating this function, ensure that its prerequisite functions have been activated and mutually exclusive functions have been deactivated, if there are any. For detailed operations, see the relevant feature documents.

Additionally, understand any potential function impacts before activating this function. To deactivate any impacted functions, follow the operations described in the relevant feature documents.

5.3.2.1 Prerequisite Functions

None

5.3.2.2 Mutually Exclusive Functions

None

5.3.2.3 Function Impacts

- Functions related to power allocation

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	LTE TDD and NR Flash Dynamic Power Sharing	LTE_NR_DYN_POWER_SHARING_SW option of the NRDUCellAlgoSwitch.DynPowerSharingSwitch parameter	<i>LTE and NR Power Allocation</i>	NR does not use the power shared by LTE in slots where UEs for which DL 256QAM is being used are scheduled.
Low-frequency TDD	LTE and NR power sharing optimization	LTE_NR_POWER_SHARING_OPT_SW option of the NRDUCellAlgoSwitch.DynPowerSharingSwitch parameter	<i>LTE and NR Power Allocation</i>	If there are UEs for which DL 256QAM is being used, the proportion of occasions where LTE and NR power sharing optimization takes effect decreases.
Low-frequency TDD	NR inter-carrier dynamic power sharing	NR_DYN_POWER_SHARING_SW option of the NRDUCellAlgoSwitch.DynPowerSharingSwitch parameter	<i>NR Inter-Carrier Dynamic Power Allocation</i>	UEs for which DL 256QAM is being used cannot benefit from NR inter-carrier dynamic power sharing.
Low-frequency TDD	NR multi-band power adaptive scheduling	NR_MULTI_BAND_PWR_ADAPT_SCH_SW option of the NRDUCellAlgoSwitch.DynPowerSharingSwitch parameter	<i>NR Inter-Carrier Dynamic Power Allocation</i>	UEs for which DL 256QAM is being used cannot benefit from NR multi-band power adaptive scheduling.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	LTE and NR multi-band power adaptive scheduling	LTE: LNR_MULTI_BAND_PWR_ADPT_SCH_SW option of the CellDynPowerSharing.DynPowerSharingSwitch parameter NR: LNR_MULTI_BAND_PWR_ADPT_SCH_SW option of the NRDUCellAlgoSwitch.DynPowerSharingSwitch parameter	<i>LTE and NR Power Allocation</i>	NR does not use the power scheduled from LTE in slots where UEs for which DL 256QAM is being used are scheduled.
Low-frequency TDD	L/NR Inter-band Deep Coordination	LTE: LNR_INTER_BAND_DEEP_COORD_SW option of the CellDynPowerSharing.DynPowerSharingSwitch parameter NR: LNR_INTER_BAND_DEEP_COORD_SW option of the NRDUCellResSshrCfg.DynPwrShrAlgoSwitch parameter	<i>LTE and NR Power Allocation</i>	NR does not use the power shared by LTE in slots where UEs for which DL 256QAM is being used are scheduled.

- Functions related to power control

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	QCI-specific downlink power control optimization	LINK_RELIABILITY_DIFF_SW option of the NRDUCellFeatureSw.QciBasedDiffServExpSw parameter DL_PWR_AGG_QCI_SW option of the gNBDUSchParamGroup.ScheduleOptSwitch parameter	<i>Service-differentiated Experience Guarantee</i>	If there are UEs for which DL 256QAM is being used, QCI-specific downlink power control optimization may take effect for a lower proportion of UEs with QCI-specific bearers.
Low-frequency TDD	RF channel intelligent shutdown	RF_SHUTDOWN_SW option of the NRDUCellAlgoSwitch.PowerSavingSwitch parameter	<i>Energy Conservation and Emission Reduction</i>	TDD 2T2R/4T4R/8T8R RF modules do not support DL 256QAM after RF channel intelligent shutdown takes effect.

- Other functions

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Low latency and high reliability	HIGH_RELIABILITY_BASIC_SW option of the NRDUCellAlgoSwitch.HighReliabilitySwitch parameter	<i>URLLC</i>	DL 256QAM does not take effect for UEs requiring low latency and high reliability.
Low-frequency TDD	Downlink power compensation for voice service UEs	VONR_DL_SERV_EXP_IMP_SW option of the NRDUCellDisch.VoiceDischSwitch parameter	<i>VoNR</i>	If there are voice service UEs for which DL 256QAM is being used, the proportion of occasions where downlink power compensation for voice service UEs takes effect may decrease.

RAT	Function Name	Function Switch	Reference	Description
High-frequency TDD	DL State Transition Buffer Volume Threshold	NRDUCellCarrMgmt.DlStateTransitBufVolThld	<i>Carrier Aggregation</i>	With DL 256QAM enabled in high-frequency TDD scenarios, a larger value of the NRDUCellCarrMgmt.DlStateTransitBufVolThld parameter results in a lower probability of SCell activation.

5.3.3 Hardware

For details about the compatibility between the hardware described in this section, see the technical specifications and hardware description of the corresponding hardware in *3900 & 5900 Series Base Station Product Documentation*.

Base Station Models

3900 and 5900 series base stations (macro base stations). 3900 series base stations must be configured with the BBU3910.

DBS3900 LampSite and DBS5900 LampSite. DBS3900 LampSite must be configured with the BBU3910. In high-frequency TDD, LampSite base stations do not support this function.

Boards

All NR-capable main control boards and baseband processing units support this function. For details, see the BBU technical specifications in *3900 & 5900 Series Base Station Product Documentation*.

RF Modules

Low-frequency scenarios: This function does not depend on RF modules.

High-frequency TDD scenarios: All NR TDD-capable AAUs that work in high frequency bands support this function. For details, see the technical specifications of AAUs in *3900 & 5900 Series Base Station Product Documentation*.

5.3.4 Others

- The value of the **pdsch-256QAM-FR1** field in the *Phy-ParametersFR1* information element (IE) in the UE capability information must be **supported** in low-frequency scenarios.
- The value of the **pdsch-256QAM-FR2** field in the *Phy-ParametersFR2* IE in the UE capability information must be **supported** in high-frequency TDD.
- Cells must not be supplementary uplink (SUL) cells. This is because SUL cells do not have downlink channels and do not support DL 256QAM or "adaptation between DL 256QAM MCS index table and DL 64QAM MCS index table".

- Neither DL 256QAM nor "adaptation between DL 256QAM MCS index table and DL 64QAM MCS index table" is supported when NR RF signals are input into LampSite base stations.

5.4 Operation and Maintenance

5.4.1 Data Configuration

5.4.1.1 Data Preparation

[Table 5-1](#) describes the parameters used for function activation. No parameters are involved in function optimization. This section does not describe parameters related to cell establishment.

Table 5-1 Parameters used for activation

Parameter Name	Parameter ID	Setting Notes
Downlink 256QAM Switch	NRDUCellAlgoSwitch.Dl256QamSwitch	Turn on this switch.

5.4.1.2 Using MML Commands

Before using MML commands, refer to [5.3 Requirements](#) and complete the parameter configurations for related functions based on the impact, dependency, and mutually exclusive relationships between the functions, as well as the actual network scenario. In addition, ensure that the license, hardware, networking, and cell requirements are met.

Before using MML commands, confirm the related parameter modification precautions by referring to the fields including "Service Interrupted After Modification" and "Caution" for the corresponding parameters in the parameter reference.

Activation Command Examples

```
//Enabling DL 256QAM
MOD NRDUCELLALGOSWITCH: NrDuCellId=0, Dl256QamSwitch=ON;
```

Deactivation Command Examples

```
//Disabling DL 256QAM
MOD NRDUCELLALGOSWITCH: NrDuCellId=0, Dl256QamSwitch=OFF;
```

5.4.1.3 Using the MAE-Deployment

For detailed operations, see Feature Configuration Using the MAE-Deployment.

 NOTE

Due to differences between OSS versions, the GUIs in the interactive video may differ from the actual OSS GUIs and are thus for reference only.

5.4.2 Activation Verification

Observe the values of the counters related to DL 256QAM. If any of the counters returns a non-zero value, DL 256QAM has taken effect.

- **N.DL.SCH.256QAM.TB**
- **N.DL.SCH.256QAM.TB.Retrans**
- **N.DL.SCH.256QAM.ErrTB.Rbler**

5.4.3 Network Monitoring

The indicators listed in [5.2 Network Analysis](#) can be used to monitor the benefits and impacts of this function.

6 DL 1024QAM

This function is supported in low frequency bands, but not in high frequency bands.

6.1 Principles

6.1.1 DL 1024QAM

DL 1024QAM results in a higher spectral efficiency than basic downlink modulation schemes and DL 256QAM, and therefore delivers a better downlink user experience.

After DL 1024QAM is enabled, the gNodeB automatically uses the DL 1024QAM MCS index table for UEs that support both DL 1024QAM and DL 256QAM. An appropriate modulation scheme can then be selected from this table according to the downlink channel quality of each UE:

- If the downlink channel quality of a UE is favorable, DL 1024QAM can be used for the UE.
- If the downlink channel quality of a UE is poor, a modulation scheme with a lower order than DL 1024QAM is used for the UE.

DL 1024QAM is controlled by the **DL_1024QAM_SW** option of the **NRDUCellDlAmc.DlModulationSchemeSw** parameter.

- Once this option is changed from deselected to selected, the DL 1024QAM MCS index table is immediately used for newly admitted UEs but is not immediately used for existing online UEs.
- Once this option is changed from selected to deselected, the DL 1024QAM MCS index table is forbidden for newly admitted UEs but is not immediately changed to another table for existing online UEs for which this table is being used.

DL 1024QAM requires DL 256QAM (controlled by **NRDUCellAlgoSwitch.Dl256QamSwitch**) to be enabled.

 NOTE

For details about modulation schemes, see [3.1 Modulation Schemes](#). For details about the DL 1024QAM MCS index table, see [3.2 MCS](#).

6.1.2 Adaptation Between DL 1024QAM MCS Index Table and DL 256QAM MCS Index Table and Between DL 256QAM MCS Index Table and DL 64QAM MCS Index Table

Downlink MCS index table adaptation is supported for high-order modulation schemes. This function enables the optimal MCS index table to be selected adaptively based on channel quality changes to improve user experience.

"Adaptation between DL 1024QAM MCS index table and DL 256QAM MCS index table and between DL 256QAM MCS index table and DL 64QAM MCS index table" is a subfunction of downlink MCS index table adaptation. Downlink MCS index table adaptation is controlled by the **DL_MCS_TABLE_ADAPT_SW** option of the **NRDUCellPdsch.DLinkAdaptAlgoSwitch** parameter. For details, see [7 Downlink MCS Index Table Adaptation](#).

6.2 Network Analysis

6.2.1 Benefits

Compared with downlink basic modulation schemes and DL 256QAM, DL 1024QAM improves downlink spectral efficiency for CCUs, and under the same channel conditions, provides the following benefits:

- Given more traffic volume, the average downlink throughput of CCUs and the average downlink cell throughput increase, and downlink data packets in the RLC buffer wait for a shorter time to be scheduled.
- Given the same or lower traffic volume, the downlink PRB usage decreases.

The degree of benefits from DL 1024QAM varies according to channel conditions. When channel conditions are favorable (for example, when the proportion of occasions where index 27 in the DL 256QAM MCS index table is selected exceeds 30% within 24 hours), DL 1024QAM can increase the **Cell Downlink Average Throughput (DU)** and **User Downlink Average Throughput (DU)** by up to 23%. In other environments or if the proportion of DL 1024QAM-capable UEs is small, this function offers fewer benefits.

When DL 1024QAM is enabled, you are advised to also enable power reduction for CCUs (detailed in *Power Control*) to ensure the benefits of DL 1024QAM.

 NOTE

Proportion of occasions where index 27 in the DL 256QAM MCS index table is selected =
(N.ChMeas.RB.PDSCH.256QAM.RANK1.MCS.27 +
N.ChMeas.RB.PDSCH.256QAM.RANK2.MCS.27 +
N.ChMeas.RB.PDSCH.256QAM.RANK3.MCS.27 +
N.ChMeas.RB.PDSCH.256QAM.RANK4.MCS.27)/(Sum of
N.ChMeas.RB.PDSCH.64QAM.RANK1.MCS.0 to
N.ChMeas.RB.PDSCH.64QAM.RANK4.MCS.28 + Sum of
N.ChMeas.RB.PDSCH.256QAM.RANK1.MCS.0 to
N.ChMeas.RB.PDSCH.256QAM.RANK4.MCS.27)

6.2.2 Impacts

The impacts between this function and other functions are described in [6.3.2.3 Function Impacts](#).

This function has the following network impacts:

- Higher-order modulation schemes deliver higher spectral efficiency and boost system capacity. However, they are more complicated to implement, provide narrower coverage, and are less resistant to interference while leading to slightly increased retransmission rates.
- Given the same MCS index, more data can be transmitted by the gNodeB with the DL 1024QAM MCS index table than with the DL 256QAM MCS index table at the cell center. Therefore, given the same amount of data to be transmitted, the average MCS index will decrease slightly after DL 1024QAM is enabled.
- Depending on UEs' weight types and channel conditions, the average number of layers when DL 1024QAM is used is likely to decrease compared to the average number when DL 256QAM is used.
- Depending on UEs' service characteristics (such as data packets' interval and size), the number of scheduling times when DL 1024QAM is used will differ from that when DL 256QAM is used.

6.3 Requirements

6.3.1 Licenses

Feature ID	Feature Name	Model	Sales Unit
FOFD-081237	DL 1024QAM	NR0S00KQAM00	Per cell

 NOTE

The insufficiency of certain feature licenses affects cell activation. For details, see the **Is Cell Activation Affected by License Insufficiency** column in *License Control Item Lists*.

6.3.2 Functions

Before activating this function, ensure that its prerequisite functions have been activated and mutually exclusive functions have been deactivated, if there are any. For detailed operations, see the relevant feature documents.

Additionally, understand any potential function impacts before activating this function. To deactivate any impacted functions, follow the operations described in the relevant feature documents.

6.3.2.1 Prerequisite Functions

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	DL 256QAM	NRDUCellAlgoSwitch.DL256QamSwitch	<i>Modulation Schemes</i>	Enabling DL 1024QAM requires DL 256QAM to be enabled.

6.3.2.2 Mutually Exclusive Functions

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Cell deployment in IAB mode	NRDUCell.DeployPosition set to DEPLOY_IAB	None	If DL 1024QAM is enabled, the NRDUCell.DeployPosition parameter cannot be set to DEPLOY_IAB .

6.3.2.3 Function Impacts

- Functions related to power allocation

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	LTE TDD and NR Flash Dynamic Power Sharing	LTE_NR_DYN_POWER_SHARING_SW option of the NRDUCellAlgoSwitch.DynPowerSharingSwitch parameter	<i>LTE and NR Power Allocation</i>	If there are UEs for which DL 1024QAM is being used, the proportion of occasions where LTE TDD and NR Flash Dynamic Power Sharing takes effect decreases.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	LTE and NR power sharing optimization	LTE_NR_POWER_SHARING_OPTION_SW option of the NRDUCellAlgoSwitch.DynPowerSharingSwitch parameter	<i>LTE and NR Power Allocation</i>	If there are UEs for which DL 1024QAM is being used, the proportion of occasions where LTE and NR power sharing optimization takes effect decreases.
Low-frequency TDD	NR inter-sector power sharing	NR_INTER_SECTOR_POWER_SHARING_SW option of the NRDUCellAlgoSwitch.DynPowerSharingSwitch parameter	None	If there are UEs for which DL 1024QAM is being used, the proportion of occasions where NR inter-sector power sharing takes effect decreases.
Low-frequency TDD	NR inter-carrier dynamic power sharing	NR_DYN_POWER_SHARING_SW option of the NRDUCellAlgoSwitch.DynPowerSharingSwitch parameter	<i>NR Inter-Carrier Dynamic Power Allocation</i>	If there are UEs for which DL 1024QAM is being used, the proportion of occasions where NR inter-carrier dynamic power sharing takes effect decreases.
Low-frequency TDD	NR multi-band power adaptive scheduling	NR_MULTI_BAND_PWR_ADAPT_SCH_SW option of the NRDUCellAlgoSwitch.DynPowerSharingSwitch parameter	<i>NR Inter-Carrier Dynamic Power Allocation</i>	If there are UEs for which DL 1024QAM is being used, the gains provided by NR multi-band power adaptive scheduling will decrease.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	LTE and NR multi-band power adaptive scheduling	LTE: LNR_MULTI_BAND_PWR_ADPT_SCH_SW option of the CellDynPowerSharing.DynamicPowerSharingSwitch parameter NR: LNR_MULTI_BAND_PWR_ADPT_SCH_SW option of the NRDUCellAlgorithmSwitch.DynPowerSharingSwitch parameter	<i>LTE and NR Power Allocation</i>	If there are UEs for which DL 1024QAM is being used, the gains provided by LTE and NR multi-band power adaptive scheduling will decrease.
Low-frequency TDD	L/NR Inter-band Deep Coordination	LTE: LNR_INTER_BAND_DEEP_COORD_SW option of the CellDynPowerSharing.DynamicPowerSharingSwitch parameter NR: LNR_INTER_BAND_DEEP_COORD_SW option of the NRDUCellResSshrCfg.DynPwrShrAlgoSwitch parameter	<i>LTE and NR Power Allocation</i>	If there are UEs for which DL 1024QAM is being used, L/NR Inter-band Deep Coordination and LTE and NR power sharing optimization will take effect in a lower proportion of cases.

- Functions related to power control

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	QCI-specific downlink power control optimization	LINK_RELIABILITY_DIFF_SW option of the NRDUCellFeatureSw.QciBasedDiffServExpSw parameter DL_PWR_AGG_QCI_SW option of the gNBDUSchParamGroup.ScheduleOptSwitch parameter	<i>Service-differentiated Experience Guarantee</i>	If there are UEs for which DL 1024QAM is being used, QCI-specific downlink power control optimization may take effect for a lower proportion of UEs with QCI-specific bearers.
Low-frequency TDD	Power aggregation in unoccupied bandwidths	NRDUCellChnPower.MaxPdschConvpwrOffset	<i>Power Control</i>	If there are UEs for which DL 1024QAM is being used, the proportion of occasions where power aggregation in unoccupied bandwidths takes effect may decrease.
Low-frequency TDD	RF channel intelligent shutdown	RF_SHUTDOWN_SW option of the NRDUCellAlgoSwitch.PowerSavingSwitch parameter	<i>Energy Conservation and Emission Reduction</i>	If RF channel intelligent shutdown takes effect, the proportion of occasions where DL 1024QAM takes effect decreases.

• Other functions

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Downlink power compensation for voice service UEs	VONR_DL_SERV_EXP_IMP_SW option of the NRDUCellIDSchVoiceDISchSwitch parameter	<i>VoNR</i>	If there are voice service UEs for which DL 1024QAM is being used, the proportion of occasions where downlink power compensation for voice service UEs takes effect may decrease.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Low latency and high reliability	HIGH_RELIABILITY_BASIC_SW option of the NRDUCellAlgoSwitch.HighReliabilitySwitch parameter	<i>URLLC</i>	DL 1024QAM does not take effect for UEs requiring low latency and high reliability.
Low-frequency TDD	Intelligent MCS optimization for downlink SU-MIMO	DL_SU_MCS_INTEL_OPT_SW option of the NRDUCellAlgoSwitch.AiAmcAlgoSwitch parameter	<i>Massive MIMO AHR (Low-Frequency TDD)</i>	Intelligent MCS optimization for downlink SU-MIMO does not take effect for UEs for which the DL 1024QAM MCS index table is selected.

6.3.3 Hardware

For details about the compatibility between the hardware described in this section, see the technical specifications and hardware description of the corresponding hardware in *3900 & 5900 Series Base Station Product Documentation*.

Base Station Models

3900 and 5900 series base stations (macro base stations). 3900 series base stations must be configured with the BBU3910.

Boards

All NR-capable main control boards and UBBPfw or later series baseband processing units support this function. For details, see the BBU technical specifications in *3900 & 5900 Series Base Station Product Documentation*.

RF Modules

Except the following RF modules, all low-frequency NR-capable RF modules on which a single NR carrier is configured can be used.

Table 6-1 RF modules that do not support DL 1024QAM

RF Module Type	RF Module
AAU	AAU3910, AAU3911, AAU3920, AAU3940 AC, AAU3940 DC, AAU3961

RF Module Type	RF Module
RRU	RRU3262, RRU3281, RRU3832, RRU3952, RRU3952m, RRU3953, RRU3953w, RRU3958, RRU3959, RRU3959a, RRU3959w, RRU3962, RRU3965, RRU3971, RRU3971a, RRU3930E, RRU3230E

6.3.4 Others

- According to section 4.2.7.2 "BandNR parameters" in 3GPP TS 38.306 V17.2.0, UEs must meet the following conditions:
 - The value of at least one of the **pdsch-1024QAM-FR1-r17** and **pdsch-1024QAM-2MIMO-FR1-r17** fields in the *BandNR* IE in the UE capability information is **supported**.
 - The value of the **pdsch-256QAM-FR1** field in the *Phy-ParametersFR1* IE in the UE capability information is **supported**.
- Cells must not be SUL cells. This is because SUL cells do not have downlink channels and do not support DL 1024QAM or "adaptation between DL 1024QAM MCS index table and DL 256QAM MCS index table and between DL 256QAM MCS index table and DL 64QAM MCS index table".
- If an **NRDUSectorSplitTrp** MO has been configured for the TRP of a 4T4R cell (with **NRDUCellTrp.TxRxMode** set to **4T4R**) and **NRDUSectorSplitGroup.BeamDeploymentMode** in the **NRDUSectorSplitGroup** MO corresponding to this TRP is set to **MODE_0**, then the settings of the **DL_1024QAM_SW** option of **NRDUCellIDAmc.DIModulationSchemeSw** must be the same in all **NRDUCellIDAmc** MOs identified by **NRDUCellTrp.NRDUCellId** in the **NRDUCellTrp** MOs identified by **NRDUSectorSplitTrp.NRDUCellTrpId** in the **NRDUSectorSplitTrp** MOs associated with this **NRDUSectorSplitGroup** MO through **NRDUSectorSplitGroup.NrDuSectorSplitGroupId**.
- If DL 1024QAM is enabled (by selecting the **DL_1024QAM_SW** option of the **NRDUCellIDAmc.DIModulationSchemeSw** parameter), the **NRDUCellTrp.FrAndDuplexMode** parameter must be set to **FR1_TDD**.

6.4 Operation and Maintenance

6.4.1 Data Configuration

6.4.1.1 Data Preparation

Table 6-2 describes the parameters used for function activation. No parameters are involved in function optimization. This section does not describe parameters related to cell establishment.

Table 6-2 Parameters used for activation

Parameter Name	Parameter ID	Option	Setting Notes
Downlink Modulation Scheme Switch	NRDUCelIDIA mc.DlModulat ionSchemeSw	DL_1024Q AM_SW	You are advised to select this option when the radio conditions are favorable, that is, when DL 256QAM has been used to the largest extent possible (for example, the proportion of occasions where MCS 27 is selected for the downlink exceeds 30% in 24 hours). For details about the recommended scenarios, see 6.2.1 Benefits .

6.4.1.2 Using MML Commands

Before using MML commands, refer to [6.3 Requirements](#) and complete the parameter configurations for related functions based on the impact, dependency, and mutually exclusive relationships between the functions, as well as the actual network scenario. In addition, ensure that the license, hardware, networking, and cell requirements are met.

Before using MML commands, confirm the related parameter modification precautions by referring to the fields including "Service Interrupted After Modification" and "Caution" for the corresponding parameters in the parameter reference.

Activation Command Examples

```
//Enabling DL 1024QAM  
MOD NRDUCELLDLAMC: NrDuCellId=0, DlModulationSchemeSw=DL_1024QAM_SW-1;
```

Deactivation Command Examples

The following provides only deactivation command examples. You can determine whether to restore the settings of other parameters based on actual network conditions.

```
//Disabling DL 1024QAM  
MOD NRDUCELLDLAMC: NrDuCellId=0, DlModulationSchemeSw=DL_1024QAM_SW-0;
```

6.4.1.3 Using the MAE-Deployment

For detailed operations, see Feature Configuration Using the MAE-Deployment.

NOTE

Due to differences between OSS versions, the GUIs in the interactive video may differ from the actual OSS GUIs and are thus for reference only.

6.4.2 Activation Verification

Observe the values of the following counters related to DL 1024QAM. If any of these counters returns a non-zero value, DL 1024QAM has taken effect.

- **N.DL.SCH.1024QAM.TB**
- **N.DL.SCH.1024QAM.ErrTB.Ibler**
- **N.DL.SCH.1024QAM.TB.Retrans**
- **N.DL.SCH.1024QAM.ErrTB.Rbler**
- **N.Traffic.DL.SCH.1024QAM.TB.Bits**
- Counters in the function subset of Measurement of CQIs in the CQI Table with 1024QAM (ChMeas.1024QAM.CQI.DuCell)
- Counters in the function subset of Measurement per MCS Index in the MCS Index Table for PDSCH with 1024QAM (ChMeas.PDSCH.1024QAM.MCS.DuCell)

6.4.3 Network Monitoring

The indicators listed in [6.2 Network Analysis](#) can be used to monitor the benefits and impacts of this function.

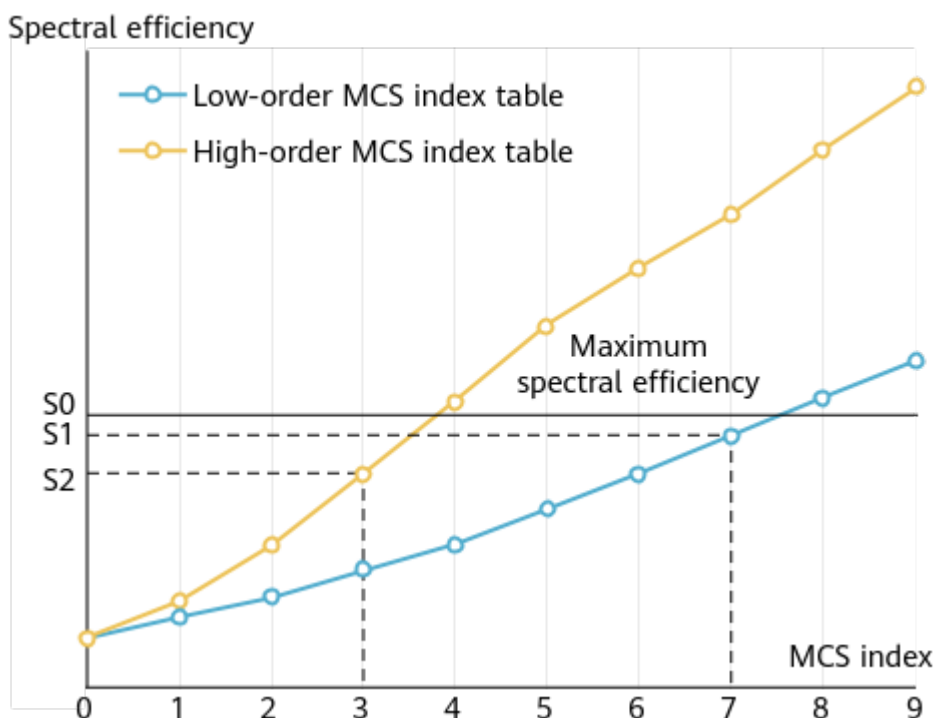
7 Downlink MCS Index Table Adaptation

7.1 Principles

The base station selects an MCS index for a UE based on the channel quality. To ensure demodulation performance, the selected MCS index must correspond to a spectral efficiency lower than or equal to a maximum spectral efficiency. The MCS index ranges from 0 to 31 across all 3GPP-defined MCS index tables.

The maximum spectral efficiency is low under poor channel conditions. Assume that the maximum spectral efficiency is S_0 , as shown in [Figure 7-1](#). As the spectral efficiency cannot exceed S_0 , MCS index 7 is selected from a low-order MCS index table to deliver spectral efficiency S_1 , and MCS index 3 is selected from a high-order MCS index table to deliver spectral efficiency S_2 . Both S_1 and S_2 are lower than S_0 , but S_1 is higher than S_2 . This means that under poor channel conditions, using a low-order MCS index table may result in a higher spectral efficiency and a better user experience than using a high-order MCS index table.

Figure 7-1 Different MCS index selections from a high-order MCS index table and a low-order MCS index table



In the downlink, there are three MCS index tables: the DL 64QAM MCS index table, DL 256QAM MCS index table, and DL 1024QAM MCS index table. If a high-order modulation scheme is to be used, a high-order MCS index table is required. Huawei provides the downlink MCS index table adaptation function. With this function, the selection of the MCS index table to be used adapts to channel condition changes to improve user experience.

Downlink MCS index table adaptation is controlled by the **DL_MCS_TABLE_ADAPT_SW** option of the **NRDUCellPdsch.DLinkAdaptAlgoSwitch** parameter. This function can be enabled only when DL 256QAM (controlled by the **NRDUCellAlgoSwitch.DL256QamSwitch** parameter) is enabled.

The downlink MCS index tables for adaptation depend on whether DL 1024QAM (controlled by the **DL_1024QAM_SW** option of the **NRDUCellDLAmc.DLModulationSchemeSw** parameter) is enabled or disabled:

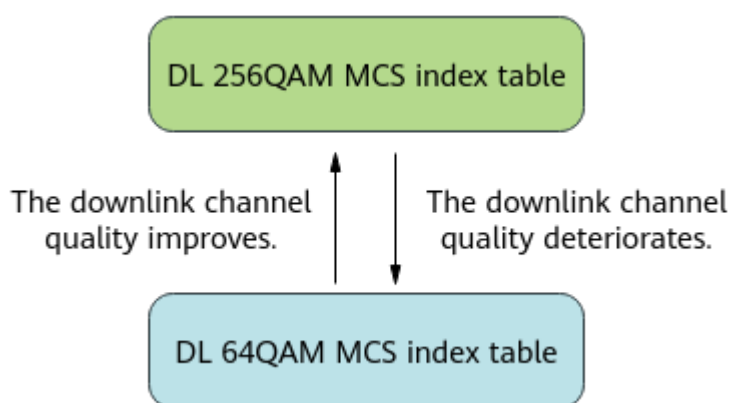
- If DL 1024QAM is disabled, "adaptation between DL 256QAM MCS index table and DL 64QAM MCS index table" takes effect. For details, see [Adaptation Between DL 256QAM MCS Index Table and DL 64QAM MCS Index Table](#) in 7.1 Principles.
- If DL 1024QAM is enabled, "adaptation between DL 1024QAM MCS index table and DL 256QAM MCS index table and between DL 256QAM MCS index table and DL 64QAM MCS index table" takes effect. Note that this adaptation function does not work between the DL 1024QAM MCS index table and the DL 64QAM MCS index table. For details, see [Adaptation Between DL 1024QAM MCS Index Table and DL 256QAM MCS Index Table and Between DL 256QAM MCS Index Table and DL 64QAM MCS Index Table](#) in 7.1 Principles.

For details about the DL 1024QAM MCS index table, DL 256QAM MCS index table, and DL 64QAM MCS index table, see section 5.1.3.1 "Modulation order and target code rate determination" of 3GPP TS 38.214 V17.1.0.

Adaptation Between DL 256QAM MCS Index Table and DL 64QAM MCS Index Table

This adaptation function takes effect if downlink MCS index table adaptation is enabled but DL 1024QAM is disabled. With this adaptation function, an MCS index table is selected according to the logic shown in [Figure 7-2](#).

Figure 7-2 Adaptation between DL 256QAM MCS index table and DL 64QAM MCS index table



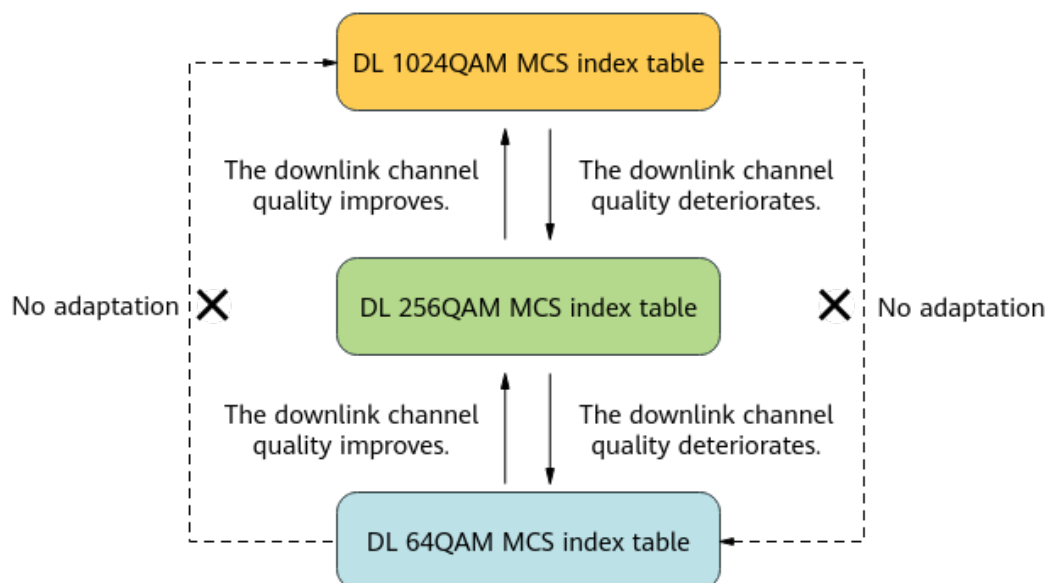
During selection of an MCS index table:

- When the downlink channel quality of a UE deteriorates, the gNodeB switches to a lower-order MCS index table for the UE. This ensures that an appropriate downlink spectral efficiency is maintained for the UE, which results in a higher downlink UE throughput.
- When the downlink channel quality of a UE improves:
 - If the UE supports a higher-order modulation scheme, the gNodeB switches to a higher-order MCS index table for the UE. This allows a higher-order modulation scheme to be used for the UE, which results in a higher downlink spectral efficiency and downlink UE throughput.
 - If the UE does not support a higher-order modulation scheme, the gNodeB continues using the current MCS index table for the UE.

Adaptation Between DL 1024QAM MCS Index Table and DL 256QAM MCS Index Table and Between DL 256QAM MCS Index Table and DL 64QAM MCS Index Table

This adaptation function takes effect if both downlink MCS index table adaptation and DL 1024QAM are enabled. Note that this adaptation function does not work between the DL 1024QAM MCS index table and the DL 64QAM MCS index table. With this adaptation function, an MCS index table is selected according to the logic shown in [Figure 7-3](#).

Figure 7-3 Adaptation between DL 1024QAM MCS index table and DL 256QAM MCS index table and between DL 256QAM MCS index table and DL 64QAM MCS index table



During selection of an MCS index table:

- When the downlink channel quality of a UE deteriorates, the gNodeB switches to a lower-order MCS index table for the UE. This ensures that an appropriate downlink spectral efficiency is maintained for the UE, which results in a higher downlink UE throughput.
- When the downlink channel quality of a UE improves:
 - If the UE supports a higher-order modulation scheme, the gNodeB switches to a higher-order MCS index table for the UE. This allows a higher-order modulation scheme to be used for the UE, which results in a higher downlink spectral efficiency and downlink UE throughput.
 - If the UE does not support a higher-order modulation scheme, the gNodeB continues using the current MCS index table for the UE.

This function is supported in low frequency bands, but not in high frequency bands.

7.2 Network Analysis

7.2.1 Benefits

Compared with the use of a fixed MCS index table, enabling downlink MCS index table adaptation provides the following benefits under the same channel conditions:

- Given more traffic volume, the average downlink throughput of CCUs and the average downlink cell throughput increase, and downlink data packets in the RLC buffer wait for a shorter time to be scheduled.
- Given the same or lower traffic volume, the downlink PRB usage decreases.

The degree of benefits from downlink MCS index table adaptation varies according to channel conditions.

- Adaptation between DL 256QAM MCS index table and DL 64QAM MCS index table

When channel conditions are complicated (for example, when the proportion of occasions where index 28 in the DL 64QAM MCS index table is selected exceeds 10% and the proportion of occasions where indexes 0 to 15 in the DL 64QAM MCS index table are selected exceeds 30% within 24 hours), this adaptation function can increase the **Cell Downlink Average Throughput (DU)** and **User Downlink Average Throughput (DU)** by up to 30%. Otherwise, the benefits are smaller.

 NOTE

Proportion of occasions where index 28 in the DL 64QAM MCS index table is selected
= $\frac{\text{N.ChMeas.PDSCH.MCS.28}}{\text{Sum of N.ChMeas.PDSCH.MCS.0 to N.ChMeas.PDSCH.MCS.28}}$

Proportion of occasions where indexes 0 to 15 in the DL 64QAM MCS index table are selected = $\frac{\text{Sum of N.ChMeas.PDSCH.MCS.0 to N.ChMeas.PDSCH.MCS.15}}{\text{Sum of N.ChMeas.PDSCH.MCS.0 to N.ChMeas.PDSCH.MCS.28}}$

- Adaptation between DL 1024QAM MCS index table and DL 256QAM MCS index table and between DL 256QAM MCS index table and DL 64QAM MCS index table

When channel conditions are favorable (for example, when the proportion of occasions where index 27 in the DL 256QAM MCS index table is selected exceeds 30% within 24 hours), this adaptation function can increase the **Cell Downlink Average Throughput (DU)** and **User Downlink Average Throughput (DU)** by up to 23%. In other environments or if the proportion of DL 1024QAM-capable UEs is small, this function offers fewer benefits.

When DL 1024QAM is enabled, you are advised to also enable power reduction for CCUs (detailed in *Power Control*) to ensure the benefits of DL 1024QAM.

 NOTE

Proportion of occasions where index 27 in the DL 256QAM MCS index table is selected
= $\frac{(\text{N.ChMeas.RB.PDSCH.256QAM.RANK1.MCS.27} + \text{N.ChMeas.RB.PDSCH.256QAM.RANK2.MCS.27} + \text{N.ChMeas.RB.PDSCH.256QAM.RANK3.MCS.27} + \text{N.ChMeas.RB.PDSCH.256QAM.RANK4.MCS.27})}{(\text{Sum of N.ChMeas.RB.PDSCH.64QAM.RANK1.MCS.0 to N.ChMeas.RB.PDSCH.64QAM.RANK4.MCS.28} + \text{Sum of N.ChMeas.RB.PDSCH.256QAM.RANK1.MCS.0 to N.ChMeas.RB.PDSCH.256QAM.RANK4.MCS.27})}$

7.2.2 Impacts

The impacts between this function and other functions are described in [7.3.2.3 Function Impacts](#).

When downlink MCS index table adaptation takes effect, downlink MCS index table switching is triggered, which has the following impacts on the network:

- If the channel quality fluctuates greatly and ping-pong switching occurs, the signaling overhead will increase.

- As MCS index table switching requires RRC reconfiguration, the values of the counters related to downlink UE throughput distribution at the RLC layer (indicated by **N.Thp.DL.Samp.Index0** to **N.Thp.DL.Samp.Index14**) may change. Specifically, the values of the counters with relatively low indexes increase and those with relatively high indexes decrease.
- The values of the following indicators change, depending on whether UEs are located near the cell center, at a medium distance from the cell center, or at the cell edge: **Downlink Resource Block Utilizing Rate (DU)**, **Cell Downlink Average Throughput (DU)**, **User Downlink Average Throughput (DU)**, **N.CCE.Used.Avg**, **N.CCE.Avail.Avg**, **N.PRB.DL.DrbUsed.Avg**, average downlink MCS index, average downlink rank, PDCCH CCE usage, indicators related to downlink scheduling, and indicators related to downlink coverage.

7.3 Requirements

7.3.1 Licenses

- Adaptation between DL 256QAM MCS index table and DL 64QAM MCS index table

Feature ID	Feature Name	Model	Sales Unit
FOFD-010050	DL 256QAM	NR0S0DUQAM00	Per cell

- Adaptation between DL 1024QAM MCS index table and DL 256QAM MCS index table and between DL 256QAM MCS index table and DL 64QAM MCS index table

Feature ID	Feature Name	Model	Sales Unit
FOFD-010050	DL 256QAM	NR0S0DUQAM00	Per cell
FOFD-081237	DL 1024QAM	NR0S00KQAM00	Per cell

NOTE

The insufficiency of certain feature licenses affects cell activation. For details, see the **Is Cell Activation Affected by License Insufficiency** column in *License Control Item Lists*.

7.3.2 Functions

Before activating this function, ensure that its prerequisite functions have been activated and mutually exclusive functions have been deactivated, if there are any. For detailed operations, see the relevant feature documents.

Additionally, understand any potential function impacts before activating this function. To deactivate any impacted functions, follow the operations described in the relevant feature documents.

7.3.2.1 Prerequisite Functions

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	DL 256QAM	NRDUCellAlgoSwitch.DL256QamSwitch	<i>Modulation Schemes</i>	Downlink MCS index table adaptation can take effect only when DL 256QAM is enabled.

7.3.2.2 Mutually Exclusive Functions

None

7.3.2.3 Function Impacts

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	High-speed Railway Superior Experience	NRDUCell.HighSpeedFlag	<i>High Speed Mobility</i>	It is recommended that "adaptation between DL 256QAM MCS index table and DL 64QAM MCS index table" be disabled for high-speed cells because UEs move at high speeds and MCS index table adaptation cannot be promptly performed to adapt to the movement. In high-speed railway scenarios, the 256QAM MCS index table is recommended (by setting the NRDUCellAlgoSwitch.DL256QamSwitch parameter to ON). Whether to optimize the configuration depends on the actual SINR and MCS index distribution.
Low-frequency TDD	MCS table control	MCS_TABLE_CTRL_SW option of the NRDUCellVoiceAlgo.VoiceUserAlgoSwitch parameter	<i>VoNR</i>	The DL 64QAM MCS index table is always used for voice service UEs when both MCS table control and "adaptation between DL 256QAM MCS index table and DL 64QAM MCS index table" are enabled.

NOTE

For the impact relationships of DL 256QAM and DL 1024QAM with other functions, see [5.3.2.3 Function Impacts](#) and [6.3.2.3 Function Impacts](#), respectively.

7.3.3 Hardware

For details about the compatibility between the hardware described in this section, see the technical specifications and hardware description of the corresponding hardware in *3900 & 5900 Series Base Station Product Documentation*.

Base Station Models

- Adaptation between DL 256QAM MCS index table and DL 64QAM MCS index table
3900 and 5900 series base stations (macro base stations). 3900 series base stations must be configured with the BBU3910.
DBS3900 LampSite and DBS5900 LampSite. DBS3900 LampSite must be configured with the BBU3910. In high-frequency TDD, LampSite base stations do not support this function.
- Adaptation between DL 1024QAM MCS index table and DL 256QAM MCS index table and between DL 256QAM MCS index table and DL 64QAM MCS index table
3900 and 5900 series base stations (macro base stations). 3900 series base stations must be configured with the BBU3910.

Boards

- Adaptation between DL 256QAM MCS index table and DL 64QAM MCS index table
All NR-capable main control boards and baseband processing units support this function. For details, see the BBU technical specifications in *3900 & 5900 Series Base Station Product Documentation*.
- Adaptation between DL 1024QAM MCS index table and DL 256QAM MCS index table and between DL 256QAM MCS index table and DL 64QAM MCS index table
All NR-capable main control boards and UBBPfw or later series baseband processing units support this function. For details, see the BBU technical specifications in *3900 & 5900 Series Base Station Product Documentation*.

RF Modules

- Adaptation between DL 256QAM MCS index table and DL 64QAM MCS index table
This function does not depend on RF modules.
- Adaptation between DL 1024QAM MCS index table and DL 256QAM MCS index table and between DL 256QAM MCS index table and DL 64QAM MCS index table
Except the following RF modules, all low-frequency NR-capable RF modules on which a single NR carrier is configured can be used.

Table 7-1 RF modules that do not support DL 1024QAM

RF Module Type	RF Module
AAU	AAU3910, AAU3911, AAU3920, AAU3940 AC, AAU3940 DC, AAU3961
RRU	RRU3262, RRU3281, RRU3832, RRU3952, RRU3952m, RRU3953, RRU3953w, RRU3958, RRU3959, RRU3959a, RRU3959w, RRU3962, RRU3965, RRU3971, RRU3971a, RRU3930E, RRU3230E

7.3.4 Others

- Adaptation between DL 256QAM MCS index table and DL 64QAM MCS index table
 - The value of the **pdsch-256QAM-FR1** field in the *Phy-ParametersFR1* IE in the UE capability information must be **supported** in low-frequency scenarios.
 - The value of the **pdsch-256QAM-FR2** field in the *Phy-ParametersFR2* IE in the UE capability information must be **supported** in high-frequency TDD.
 - Cells must not be SUL cells. This is because SUL cells do not have downlink channels and do not support DL 256QAM or "adaptation between DL 256QAM MCS index table and DL 64QAM MCS index table".
 - Neither DL 256QAM nor "adaptation between DL 256QAM MCS index table and DL 64QAM MCS index table" is supported when NR RF signals are input into LampSite base stations.
- Adaptation between DL 1024QAM MCS index table and DL 256QAM MCS index table and between DL 256QAM MCS index table and DL 64QAM MCS index table
 - According to section 4.2.7.2 "BandNR parameters" in 3GPP TS 38.306 V17.2.0, UEs must meet the following conditions:
 - The value of at least one of the **pdsch-1024QAM-FR1-r17** and **pdsch-1024QAM-2MIMO-FR1-r17** fields in the *BandNR* IE in the UE capability information is **supported**.
 - The value of the **pdsch-256QAM-FR1** field in the *Phy-ParametersFR1* IE in the UE capability information is **supported**.
 - Cells must not be SUL cells. This is because SUL cells do not have downlink channels and do not support DL 1024QAM or "adaptation between DL 1024QAM MCS index table and DL 256QAM MCS index table and between DL 256QAM MCS index table and DL 64QAM MCS index table".

7.4 Operation and Maintenance

7.4.1 Data Configuration

7.4.1.1 Data Preparation

Table 7-2 describes the parameters used for function activation. No parameters are involved in function optimization. This section does not describe parameters related to cell establishment.

Table 7-2 Parameters used for activation

Parameter Name	Parameter ID	Option	Setting Notes
Downlink 256QAM Switch	NRDUCellAlgoSwitch.DL256QamSwitch	None	Turn on this switch.
Downlink Modulation Scheme Switch	NRDUCellDIAmc.DIModulationSchemeSw	DL_1024QAM_SW	Select this option.
DL Link Adaptation Algorithm Switch	NRDUCellPdsc h.DLLinkAdaptAlgoSwitch	DL_MCS_TABLE_ADAPT_SW	You are advised to select this option, which specifies whether to enable downlink MCS index table adaptation.

7.4.1.2 Using MML Commands

Before using MML commands, refer to **7.3 Requirements** and complete the parameter configurations for related functions based on the impact, dependency, and mutually exclusive relationships between the functions, as well as the actual network scenario. In addition, ensure that the license, hardware, networking, and cell requirements are met.

Before using MML commands, confirm the related parameter modification precautions by referring to the fields including "Service Interrupted After Modification" and "Caution" for the corresponding parameters in the parameter reference.

Activation Command Examples

- Adaptation between DL 256QAM MCS index table and DL 64QAM MCS index table
//Enabling both DL 256QAM and downlink MCS index table adaptation and disabling DL 1024QAM to enable "adaptation between DL 256QAM MCS index table and DL 64QAM MCS index table"
//Enabling DL 256QAM
MOD NRDUCCELLALGOSWITCH: NrDuCellId=0, DL256QamSwitch=ON;
//Enabling downlink MCS index table adaptation
MOD NRDUCCELLPDSCH: NrDuCellId=0, DLLinkAdaptAlgoSwitch=DL_MCS_TABLE_ADAPT_SW-1;
- Adaptation between DL 1024QAM MCS index table and DL 256QAM MCS index table and between DL 256QAM MCS index table and DL 64QAM MCS index table
//Enabling DL 256QAM, DL 1024QAM, and downlink MCS index table adaptation to enable "adaptation between DL 1024QAM MCS index table and DL 256QAM MCS index table and between

```
DL 256QAM MCS index table and DL 64QAM MCS index table"
//Enabling DL 256QAM
MOD NRDUCELLALGOSWITCH: NrDuCellId=0, DL256QamSwitch=ON;
//Enabling DL 1024QAM
MOD NRDUCELLDLAMC: NrDuCellId=0, DLModulationSchemeSw=DL_1024QAM_SW-1;
//Enabling downlink MCS index table adaptation
MOD NRDUCELLPDSCH: NrDuCellId=0, DLLinkAdaptAlgoSwitch=DL_MCS_TABLE_ADAPT_SW-1;
```

Deactivation Command Examples

The following provides only deactivation command examples. You can determine whether to restore the settings of other parameters based on actual network conditions.

- Adaptation between DL 256QAM MCS index table and DL 64QAM MCS index table

```
//Disabling "adaptation between DL 256QAM MCS index table and DL 64QAM MCS index table"
MOD NRDUCELLPDSCH: NrDuCellId=0, DLLinkAdaptAlgoSwitch=DL_MCS_TABLE_ADAPT_SW-0;
//Disabling DL 256QAM
MOD NRDUCELLALGOSWITCH: NrDuCellId=0, DL256QamSwitch=OFF;
```

- Adaptation between DL 1024QAM MCS index table and DL 256QAM MCS index table and between DL 256QAM MCS index table and DL 64QAM MCS index table

```
//Disabling "adaptation between DL 1024QAM MCS index table and DL 256QAM MCS index table
and between DL 256QAM MCS index table and DL 64QAM MCS index table"
MOD NRDUCELLPDSCH: NrDuCellId=0, DLLinkAdaptAlgoSwitch=DL_MCS_TABLE_ADAPT_SW-0;
//Disabling DL 1024QAM
MOD NRDUCELLDLAMC: NrDuCellId=0, DLModulationSchemeSw=DL_1024QAM_SW-0;
//Disabling DL 256QAM
MOD NRDUCELLALGOSWITCH: NrDuCellId=0, DL256QamSwitch=OFF;
```

7.4.1.3 Using the MAE-Deployment

For detailed operations, see Feature Configuration Using the MAE-Deployment.

NOTE

Due to differences between OSS versions, the GUIs in the interactive video may differ from the actual OSS GUIs and are thus for reference only.

7.4.2 Activation Verification

When DL 256QAM and downlink MCS index table adaptation are both enabled and DL 1024QAM is disabled, "adaptation between DL 256QAM MCS index table and DL 64QAM MCS index table" is enabled. In this case, observe the values of the following counters. If any of the counters returns a non-zero value, this adaptation function has taken effect.

- **N.DL.SCH.256QAM.TB**
- **N.DL.SCH.256QAM.TB.Retrans**
- **N.DL.SCH.256QAM.ErrTB.Rbler**

When DL 256QAM, DL 1024QAM, and downlink MCS index table adaptation are all enabled, "adaptation between DL 1024QAM MCS index table and DL 256QAM MCS index table and between DL 256QAM MCS index table and DL 64QAM MCS index table" is enabled. In this case, observe the values of the following counters. If any of the counters returns a non-zero value, this adaptation function has taken effect.

- **N.DL.SCH.1024QAM.TB**
- **N.DL.SCH.1024QAM.ErrTB.Ibler**
- **N.DL.SCH.1024QAM.TB.Retrans**
- **N.DL.SCH.1024QAM.ErrTB.Rbler**
- **N.Traffic.DL.SCH.1024QAM.TB.Bits**

7.4.3 Network Monitoring

Compare the total volume of downlink data sent at the MAC layer (indicated by **N.ThpVol.DL.Cell**) or the downlink PRB usage (equal to **N.PRB.DL.Used.Avg**/**N.PRB.DL.Avail.Avg**) before and after this function is enabled to evaluate the gains.

8 UL 256QAM

8.1 Principles

The UL 256QAM feature includes the following functions:

- UL 256QAM
- Adaptation between UL 256QAM MCS index table and UL 64QAM MCS index table

These two functions cannot be both enabled.

8.1.1 UL 256QAM

UL 256QAM is an enhancement to uplink basic modulation schemes. This function is enabled when the **NRDUCellAlgoSwitch.Ul256QamSwitch** parameter is set to **UL_256QAM_FIXED**. With this function:

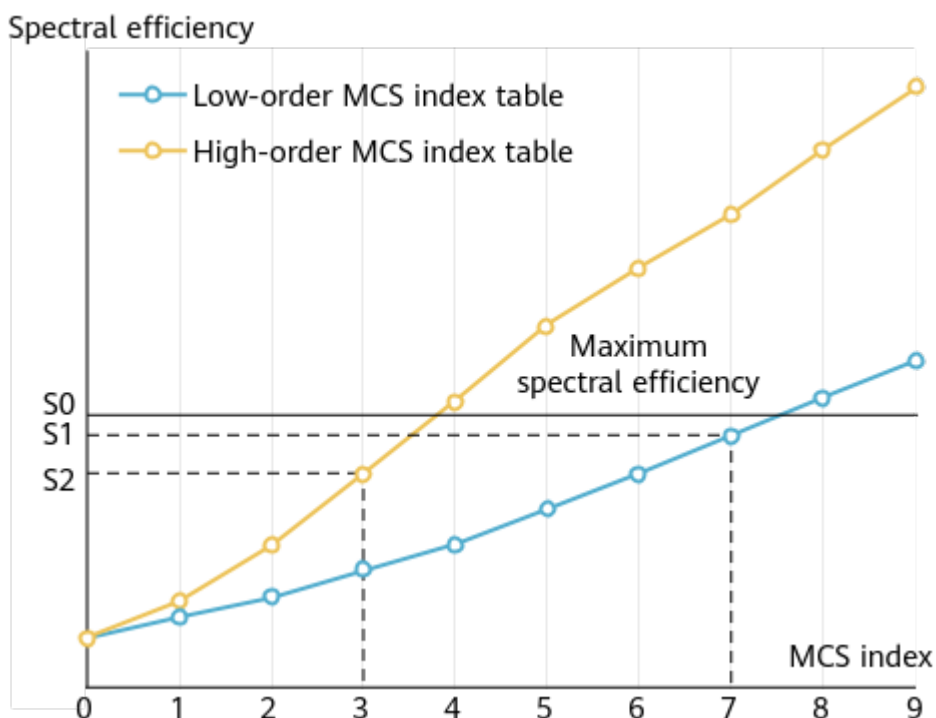
- The UL 256QAM MCS index table is used for UEs that support UL 256QAM.
- UL 256QAM is used for UL 256QAM-capable UEs under favorable uplink channel conditions.

8.1.2 Uplink MCS Index Table Adaptation

The base station selects an MCS index for a UE based on the channel quality. To ensure demodulation performance, the selected MCS index must correspond to a spectral efficiency lower than or equal to a maximum spectral efficiency. The MCS index ranges from 0 to 31 across all 3GPP-defined MCS index tables.

The maximum spectral efficiency is low under poor channel conditions. Assume that the maximum spectral efficiency is S_0 , as shown in [Figure 8-1](#). As the spectral efficiency cannot exceed S_0 , MCS index 7 is selected from a low-order MCS index table to deliver spectral efficiency S_1 , and MCS index 3 is selected from a high-order MCS index table to deliver spectral efficiency S_2 . Both S_1 and S_2 are lower than S_0 , but S_1 is higher than S_2 . This means that under poor channel conditions, using a low-order MCS index table may result in a higher spectral efficiency and a better user experience than using a high-order MCS index table.

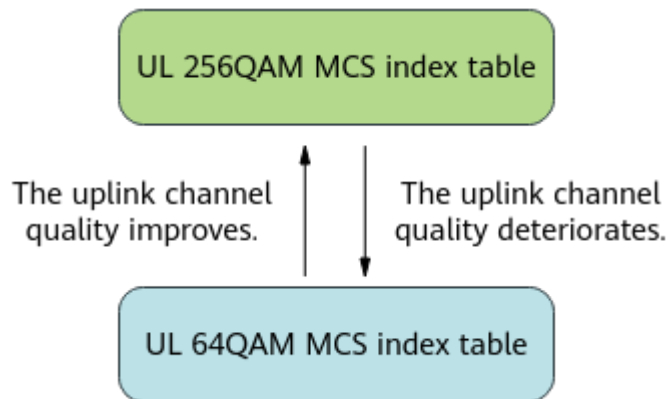
Figure 8-1 Different MCS index selections from a high-order MCS index table and a low-order MCS index table



In the uplink, "adaptation between UL 256QAM MCS index table and UL 64QAM MCS index table" is supported. This function is enabled when the **NRDUCellAlgoSwitch.Ul256QamSwitch** parameter is set to **UL_256QAM_ADAPT**. With this function, the UL 256QAM MCS index table or UL 64QAM MCS index table can be adaptively selected, as shown in [Figure 8-2](#). Specifically:

- When the uplink channel quality of a UE improves, the UL 256QAM MCS index table is used for the UE. This allows 256QAM to be used for the UE, which results in a higher uplink spectral efficiency and uplink UE throughput.
- When the uplink channel quality of a UE deteriorates, the UL 64QAM MCS index table is used for the UE. This ensures that an appropriate uplink spectral efficiency is maintained for the UE, which results in a higher uplink UE throughput.

Figure 8-2 Uplink MCS index table adaptation



For details about the UL 256QAM MCS index table and UL 64QAM MCS index table, see section 6.1.4.1 "Modulation order and target code rate determination" of 3GPP TS 38.214 V15.4.0.

"Adaptation between UL 256QAM MCS index table and UL 64QAM MCS index table" can work with the following functions for better user experience and base station performance:

- **UL 256QAM-based adaptive MCS index selection and power control threshold optimization:** This function increases the probability of switching from UL 64QAM to UL 256QAM and improves the uplink data rates and spectral efficiency for CCUs. This function is controlled by the **UL_256QAM_ADAPT_POWER_OPT_SW** option of the **NRDUCellUIMc.UlMcAlgoSw** parameter.
- **Lost-UE protection against service drops:** This function reduces the service drop rate in cells by optimizing the uplink MCS index table reconfiguration procedures for lost UEs (UEs that voluntarily stop sending data on the NR PUSCH but have not been released due to timeout by the core network). It is recommended that this function be enabled for a cell that serves lost UEs when the service drop rate increases because lost UEs cannot respond to RRC connection reconfiguration messages for switching their uplink MCS index tables. This function is controlled by the **LOST_UE_DROP_PROTECT_SW** option of the **NRDUCellUIMc.UlMcPerformanceSw** parameter.

8.2 Network Analysis

8.2.1 Benefits

UL 256QAM

Compared with uplink basic modulation schemes, UL 256QAM improves uplink spectral efficiency for CCUs, and under the same channel conditions, provides the following benefits:

- Given more traffic volume, the average uplink throughput of CCUs and the average uplink cell throughput increase.

- Given the same or lower traffic volume, the uplink PRB usage decreases.

The degree of benefits from UL 256QAM varies according to channel conditions. When channel conditions are favorable (for example, when the proportion of occasions where index 28 in the UL 64QAM MCS index table is selected exceeds 30% within 24 hours), UL 256QAM can increase the **Cell Uplink Average Throughput (DU)** and **User Uplink Average Throughput (DU)** by up to 30%. Otherwise, the benefits are smaller.

 NOTE

Proportion of occasions where index 28 in the UL 64QAM MCS index table is selected = $\frac{N.ChMeas.PUSCH.MCS.28}{\text{Sum of } N.ChMeas.PUSCH.MCS.0 \text{ to } N.ChMeas.PUSCH.MCS.28}$

Adaptation Between UL 256QAM MCS Index Table and UL 64QAM MCS Index Table

Compared with the use of a fixed MCS index table, enabling "adaptation between UL 256QAM MCS index table and UL 64QAM MCS index table" provides the following benefits under the same channel conditions:

- Given more traffic volume, the average uplink throughput of CCUs and the average uplink cell throughput increase.
- Given the same or lower traffic volume, the uplink PRB usage decreases.

The degree of benefits from this adaptation function varies according to channel conditions. When channel conditions are complicated (for example, when the proportion of occasions where index 28 in the UL 64QAM MCS index table is selected exceeds 10% and the proportion of occasions where indexes 0 to 15 in the UL 64QAM MCS index table are selected exceeds 30% within 24 hours), this adaptation function can increase the **Cell Uplink Average Throughput (DU)** and **User Uplink Average Throughput (DU)** by up to 30%. Otherwise, the benefits are smaller.

 NOTE

Proportion of occasions where index 28 in the UL 64QAM MCS index table is selected = $\frac{N.ChMeas.PUSCH.MCS.28}{\text{Sum of } N.ChMeas.PUSCH.MCS.0 \text{ to } N.ChMeas.PUSCH.MCS.28}$

Proportion of occasions where indexes 0 to 15 in the UL 64QAM MCS index table are selected = $\frac{\text{Sum of } N.ChMeas.PUSCH.MCS.0 \text{ to } N.ChMeas.PUSCH.MCS.15}{\text{Sum of } N.ChMeas.PUSCH.MCS.0 \text{ to } N.ChMeas.PUSCH.MCS.28}$

UL 256QAM-based adaptive MCS index selection and power control threshold optimization increases the probability of switching from UL 64QAM to UL 256QAM and improves the average uplink throughput and spectral efficiency for CCUs.

Lost-UE protection against service drops may decrease the **Service Call Drop Rate (CU, Inactive)** when there are lost UEs in a cell.

8.2.2 Impacts

The impacts between this function and other functions are described in [8.3.2.3 Function Impacts](#).

Higher-order modulation schemes deliver higher spectral efficiency and boost system capacity. However, they are more complicated to implement, provide

narrower coverage, and are less resistant to interference while leading to slightly increased retransmission rates and service drop rates.

Given the same MCS index, more data can be transmitted by the gNodeB with the UL 256QAM MCS index table than with the UL 64QAM MCS index table in cell-center areas. Therefore, given the same amount of data to be transmitted, the average MCS index will decrease slightly after UL 256QAM is enabled.

The UE transmit power when UL 256QAM is used is likely to be lower than that when a basic modulation scheme is used.

Depending on UE's service characteristics (such as the interval and size of data packets), the CCE usage (**N.CCE.Used.Avg/N.CCE.Avail.Avg**) and number of scheduling times when UL 256QAM is used will differ from those when UL 64QAM is used.

When "adaptation between UL 256QAM MCS index table and UL 64QAM MCS index table" takes effect, switching between the two tables is triggered. If the channel quality fluctuates greatly and ping-pong switching occurs, the signaling overhead will increase.

When the **NRDUCellAlgoSwitch.Ul256QamSwitch** parameter is set to **UL_256QAM_FIXED**, the gNodeB always uses the MCS index table with 256QAM for 256QAM-capable UEs during uplink scheduling. Compared with the MCS index table without 256QAM, the MCS index table with 256QAM provides fewer available low-order spectral efficiency levels. As a result, the uplink data rates of 256QAM-capable UEs at the cell edge may decrease.

When uplink power reduction optimization is in effect and the modulation scheme is changed from UL 64QAM to UL 256QAM, the uplink BLER decreases and the uplink throughput increases. It is recommended that uplink power reduction optimization be enabled when UL 256QAM is enabled. Uplink power reduction optimization is controlled by the **UL_256QAM_MPR_OPT_SW** option of the **NRDUCellUlamc.UlAmcAlgoSw** parameter. For details about this function, see *Uplink Scheduling*.

Lost-UE protection against service drops reduces the switching between 256QAM and 64QAM index tables, which changes the proportions of cases where 256QAM and 64QAM are used in the uplink, the average uplink MCS index in a cell, and the **User Uplink Average Throughput (DU)**.

8.3 Requirements

8.3.1 Licenses

Feature ID	Feature Name	Model	Sales Unit
FOFD-030215	UL 256QAM	NR0SUL256QAM	Per cell

 NOTE

The insufficiency of certain feature licenses affects cell activation. For details, see the **Is Cell Activation Affected by License Insufficiency** column in *License Control Item Lists*.

8.3.2 Functions

Before activating this function, ensure that its prerequisite functions have been activated and mutually exclusive functions have been deactivated, if there are any. For detailed operations, see the relevant feature documents.

Additionally, understand any potential function impacts before activating this function. To deactivate any impacted functions, follow the operations described in the relevant feature documents.

8.3.2.1 Prerequisite Functions

None

8.3.2.2 Mutually Exclusive Functions

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	UL $\pi/2$ BPSK	UL_HALF_PI_BPSK_SW option of the NRDUCellAlgoSwitch.AdaptiveEdgeExpEnhSwitch parameter	<i>Turbo Uplink (Low-Frequency TDD)</i>	If the NRDUCellAlgoSwitch.Ul256QamSwitch parameter is set to UL_256QAM_FIXED , the UL_HALF_PI_BPSK_SW option of the NRDUCellAlgoSwitch.AdaptiveEdgeExpEnhSwitch parameter must be deselected for the corresponding cell.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Uplink low-PAPR DMRS	UL_LOW_PAPR_DMRS_SW option of the NRDUCellAlgoSwitch.AdaptiveEdgeExpEnhSwitch parameter	<i>Turbo Uplink (Low-Frequency TDD)</i>	If the NRDUCellAlgoSwitch.Ul256QamSwitch parameter is set to UL_256QAM_FIXED , the UL_LOW_PAPR_DMRS_SW option of the NRDUCellAlgoSwitch.AdaptiveEdgeExpEnhSwitch parameter must be deselected for the corresponding cell.
Low-frequency TDD	Cell deployment in IAB mode	NRDUCell.DeployPosition set to DEPLOY_IAB	None	For cells deployed in IAB mode, the NRDUCellAlgoSwitch.Ul256QamSwitch parameter must be set to UL_256QAM_OFF .
High-frequency TDD	None	None	None	None

8.3.2.3 Function Impacts

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	High-speed Railway Superior Experience	NRDUCell.HighSpeedFlag	<i>High Speed Mobility</i>	It is recommended that "adaptation between UL 256QAM MCS index table and UL 64QAM MCS index table" be disabled for high-speed cells because UEs move at high speeds and MCS index table adaptation cannot be promptly performed to adapt to the movement. In high-speed railway scenarios, the 256QAM MCS index table is recommended (by setting the NRDUCellAlgoSwitch.Ul256QamSwitch parameter to UL_256QAM_FIXED). In a specific scenario, whether to optimize the configuration depends on the SINR and MCS index distribution.
Low-frequency TDD	Low latency and high reliability	HIGH_RELIABILITY_BASIC_SW option of the NRDUCellAlgoSwitch.HighReliabilitySwitch parameter	<i>URLLC</i>	UL 256QAM does not take effect for UEs requiring low latency and high reliability.
Low-frequency TDD	High-reliability uplink MCS selection	UL_HIGH_RELIAB_MCS_SELECTION option of the NRDUCellUplinkMcPerformanceSwitch parameter	<i>Turbo Uplink (Low-Frequency TDD)</i>	If the UL 256QAM MCS index table is used for CEUs, RB estimation is inaccurate. If the uplink QPSK proportion increases after high-reliability uplink MCS selection is enabled, the uplink user-perceived rate of CEUs will be affected. Therefore, it is recommended that "adaptation between UL 256QAM MCS index table and UL 64QAM MCS index table" be enabled as well. (Uplink QPSK proportion = $\frac{N_{UL.SCH.QPSK.TB}}{(N_{UL.SCH.QPSK.TB} + N_{UL.SCH.16QAM.TB} + N_{UL.SCH.64QAM.TB} + N_{UL.SCH.256QAM.TB})} \times 100\%$)

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	MCS table control	MCS_TABLE_CTRL_SW option of the NRDUCellVoiceAlgo.VoiceUserAlgoSwitch parameter	<i>VoNR</i>	The UL 64QAM MCS index table is always used for voice service UEs when both MCS table control and "adaptation between UL 256QAM MCS index table and UL 64QAM MCS index table" are enabled.
Low-frequency TDD	Intra-base-station UL CoMP	INTRA_GNB_UL_COMP_SW option of the NRDUCellAlgoSwitch.CompSwitch parameter	<i>CoMP</i>	For UEs with UL 256QAM in effect, intra-base-station UL CoMP can take effect for them when the SBC_COMPATIBLE_WITH_256QAM_SW option of the NRDUCellComp.UlCompAlgoSwitch parameter is selected, and does not take effect when the option is deselected.

8.3.3 Hardware

For details about the compatibility between the hardware described in this section, see the technical specifications and hardware description of the corresponding hardware in *3900 & 5900 Series Base Station Product Documentation*.

Base Station Models

3900 and 5900 series base stations (macro base stations). 3900 series base stations must be configured with the BBU3910.

DBS3900 LampSite and DBS5900 LampSite. DBS3900 LampSite must be configured with the BBU3910.

Boards

All NR-capable main control boards and baseband processing units support this function. For details, see the BBU technical specifications in *3900 & 5900 Series Base Station Product Documentation*.

RF Modules

This function does not depend on RF modules.

8.3.4 Others

The value of the **pusch-256QAM** field in the *RF-Parameters* IE in the UE capability information must be **supported**.

8.4 Operation and Maintenance

8.4.1 Data Configuration

8.4.1.1 Data Preparation

Table 8-1 describes the parameters used for function activation. This section does not describe parameters related to cell establishment.

Table 8-1 Parameters used for activation

Parameter Name	Parameter ID	Option	Setting Notes
Uplink 256QAM Switch	NRDUCellAlgoSwitch.Ul256QamSwitch	None	Set this parameter to UL_256QAM_FIXED or UL_256QAM_ADAPT based on network conditions. When uplink power reduction optimization is in effect and the modulation scheme is changed from UL 64QAM to UL 256QAM, the uplink IBLER decreases and the uplink throughput increases. It is recommended that uplink power reduction optimization be enabled when UL 256QAM is enabled. For details about uplink power reduction optimization, see <i>Uplink Scheduling</i> .

Parameter Name	Parameter ID	Option	Setting Notes
Uplink AMC Performance Switch	NRDUCellULAmc. ULAmcPerformanceSw	LOST_UE_DROP_PROTECT_SW	It is recommended that this option be selected if there are lost UEs in a cell.
Uplink AMC Algorithm Switch	NRDUCellULAmc. ULAmcAlgoSw	UL_256QAM_ADAPT_POWER_OPT_SW	If the NRDUCellAlgoSwitch.UL256QamSwitch parameter is set to UL_256QAM_ADAPT , it is recommended that this option be selected to enable UL 256QAM-based adaptive MCS index selection and power control threshold optimization.

8.4.1.2 Using MML Commands

Before using MML commands, refer to [8.3 Requirements](#) and complete the parameter configurations for related functions based on the impact, dependency, and mutually exclusive relationships between the functions, as well as the actual network scenario. In addition, ensure that the license, hardware, networking, and cell requirements are met.

Before using MML commands, confirm the related parameter modification precautions by referring to the fields including "Service Interrupted After Modification" and "Caution" for the corresponding parameters in the parameter reference.

Activation Command Examples

```
//Enabling UL 256QAM or "adaptation between UL 256QAM MCS index table and UL 64QAM MCS index table" based on radio network conditions
//Enabling UL 256QAM
MOD NRDUCELLALGOSWITCH: NrDuCellId=0, UL256QamSwitch=UL_256QAM_FIXED;
//Enabling "adaptation between UL 256QAM MCS index table and UL 64QAM MCS index table"
MOD NRDUCELLALGOSWITCH: NrDuCellId=0, UL256QamSwitch=UL_256QAM_ADAPT;
//Enabling lost-UE protection against service drops when there are lost UEs in the cell
MOD NRDUCELLULAMC: NrDuCellId=0, ULAmcPerformanceSw=LOST_UE_DROP_PROTECT_SW-1;
//((Recommended) Enabling UL 256QAM-based adaptive MCS index selection and power control threshold optimization when "adaptation between UL 256QAM MCS index table and UL 64QAM MCS index table" is enabled
MOD NRDUCELLULAMC: NrDuCellId=0, ULAmcAlgoSw=UL_256QAM_ADAPT_POWER_OPT_SW-1;
```


Deactivation Command Examples

The following provides only deactivation command examples. You can determine whether to restore the settings of other parameters based on actual network conditions.

```
//(Optional) Disabling UL 256QAM-based adaptive MCS index selection and power control threshold optimization  
MOD NRDUCELLULAMC: NrDuCellId=0, UIAmcAlgoSw=UL_256QAM_ADAPT_POWER_OPT_SW-0;  
//(Optional) Disabling lost-UE protection against service drops  
MOD NRDUCELLULAMC: NrDuCellId=0, UIAmcPerformanceSw=LOST_UE_DROP_PROTECT_SW-0;  
//Disabling UL 256QAM and "adaptation between UL 256QAM MCS index table and UL 64QAM MCS index table"  
MOD NRDUCELLALGOSWITCH: NrDuCellId=0, UI256QamSwitch=UL_256QAM_OFF;
```

8.4.1.3 Using the MAE-Deployment

For detailed operations, see Feature Configuration Using the MAE-Deployment.

NOTE

Due to differences between OSS versions, the GUIs in the interactive video may differ from the actual OSS GUIs and are thus for reference only.

8.4.2 Activation Verification

Observe the values of the counters related to UL 256QAM. If any of the counters returns a non-zero value, UL 256QAM has taken effect.

- **N.UL.SCH.256QAM.TB**
- **N.UL.SCH.256QAM.TB.Retrans**
- **N.UL.SCH.256QAM.ErrTB.Rbler**

8.4.3 Network Monitoring

The indicators listed in [8.2 Network Analysis](#) can be used to monitor the benefits and impacts of this function.

9 Parameters

The following hyperlinked EXCEL files of parameter reference match the software version with which this document is released.

- Node Parameter Reference: contains device and transport parameters.
- Node Used Reserved Parameter List: contains the reserved parameters that are in use and those that have been disused.
- gNodeBFunction Parameter Reference: contains all parameters related to radio access functions, including air interface management, access control, mobility control, and radio resource management.
- 3900 & 5900 Series Base Station gNodeBFunction Used Reserved Parameter List: contains the reserved parameters that are in use and the reserved parameters that fall into disuse in the current R version.

NOTE

You can find the EXCEL files of parameter reference and used reserved parameter list for the software version used on the live network from the "Man-Machine Interface Reference" node in the *3900 & 5900 Series Base Station Product Documentation* delivered with that version.

FAQ 1: How do I find the parameters related to a certain feature from parameter reference?

- Step 1** Open the EXCEL file of parameter reference.
- Step 2** On the **Parameter List** sheet, filter the **Feature ID** column. Click **Text Filters** and choose **Contains**. Enter the feature ID, for example, FBFD-010011.
- Step 3** Click **OK**. All parameters related to the feature are displayed.

----End

FAQ 2: How do I find the information about a certain reserved parameter from the used reserved parameter list?

- Step 1** Open the EXCEL file of the used reserved parameter list.

Step 2 On the **Used Reserved Parameter List** sheet, use the **MO**, **Parameter ID**, and **BIT** columns to locate the reserved parameter. View its information, including the meaning, values, and impacts.

----End

10 Counters

The following hyperlinked EXCEL files of performance counter reference match the software version with which this document is released.

- Node Performance Counter Summary: contains device and transport counters.
- gNodeBFunction Performance Counter Summary: contains all counters related to radio access functions, including air interface management, access control, mobility control, and radio resource management.
- gNodeBFunction Used Reserved Counter List: contains the reserved counters that are in use and those that have been disused.

NOTE

You can find the EXCEL files of performance counter reference and used reserved counter list for the software version used on the live network from the "Man-Machine Interface Reference" node in the *3900 & 5900 Series Base Station Product Documentation* delivered with that version.

FAQ: How do I find the counters related to a certain feature from performance counter reference?

- Step 1** Open the EXCEL file of performance counter reference.
- Step 2** On the **Counter Summary(En)** sheet, filter the **Feature ID** column. Click **Text Filters** and choose **Contains**. Enter the feature ID, for example, FBFD-010011.
- Step 3** Click **OK**. All counters related to the feature are displayed.

----End

11 Glossary

For the acronyms, abbreviations, terms, and definitions, see *Glossary*.

12 Reference Documents

- *LTE and NR Power Allocation*
- *High Speed Mobility*
- *URLLC*
- *Energy Conservation and Emission Reduction*
- *Uplink Scheduling*
- *Downlink Scheduling*
- *CoMP*
- *Turbo Uplink (Low-Frequency TDD)*
- *Power Control*
- *VoNR*
- *NR Inter-Carrier Dynamic Power Allocation*
- *Carrier Aggregation*
- *Massive MIMO AHR (Low-Frequency TDD)*
- *Service-differentiated Experience Guarantee*
- *License Control Item Lists*
- 3GPP TS 38.214: "NR; Physical layer procedure for data"
- 3GPP TS 38.306: "NR; User Equipment (UE) radio access capabilities"
- 3GPP TS 38.211: "NR; Physical channels and modulation"
- 3GPP TS 38.104: "NR; Base Station (BS) radio transmission and reception"
- *Technical Specifications in 3900 & 5900 Series Base Station Product Documentation*